

		Wavelength / 10^{-11} m																					
8	(a)	There are <u>3 positively charged particles</u> <u>Applying Fleming's Left Hand Rule</u> , there are 3 lines curving to the left.	A1 M1																				
	(b)(i)	$r \propto \frac{p}{q}$ Compared to other tracks, the <u>radius of the track in Fig. 8.3 is the smallest</u> . The <u>mass of electron is small</u> . Its momentum and the radius will be small. Hence the proposal is possible.	M1 A1																				
	(b)(ii)	The electron <u>loses energy</u> as it travels through the chamber. Its <u>momentum and hence its radius decreases</u> , resulting in a spiral.	B1 B1																				
	(c)(i)	Conservation of charge gives (Charge of K^-) + (proton charge) = Total charge after collision $(-e) + (+e) = (+e) + (-e) + (+e) + q_4$ $\Rightarrow q_4 = -e$	C1 A1																				
	(c)(ii)	$r \propto \frac{p}{q}$ Particle 3 has the lowest momentum because r is smallest.	A1																				
	(d)(i)	It causes little or no ionization in the target atoms, so it leaves no track.	B1																				
	(d)(ii)	They are oppositely charged, as the paths curve in opposite directions, implying that the magnetic force points in opposite directions, OR, because they are formed from a neutral particle.	B1 B1																				
	(e)(i) 1. (e)(i) 2.	<table><tr><td>particle</td><td>$p_x / 10^{-20}$ N s</td><td>$p_y / 10^{-20}$ N s</td><td>$p_z / 10^{-20}$ N s</td><td>$E / 10^{-12}$ J</td></tr><tr><td>K^-</td><td>438.05</td><td>-13.24</td><td>0.81</td><td>1317.12</td></tr><tr><td>p</td><td>0.00</td><td>0.00</td><td>0.00</td><td>150.13</td></tr><tr><td>sum</td><td>438.05</td><td>-13.24</td><td>0.81</td><td>1467.25</td></tr></table>	particle	$p_x / 10^{-20}$ N s	$p_y / 10^{-20}$ N s	$p_z / 10^{-20}$ N s	$E / 10^{-12}$ J	K^-	438.05	-13.24	0.81	1317.12	p	0.00	0.00	0.00	150.13	sum	438.05	-13.24	0.81	1467.25	A1
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	(e)(ii)1	$p_x = (438.05 - 358.59) \times 10^{-20} = +79.46 \times 10^{-20} \text{ N s}$ $p_y = (-13.24 + 7.48) \times 10^{-20} = -5.76 \times 10^{-20} \text{ N s}$ $p_z = (0.81 + 4.02) \times 10^{-20} = +4.83 \times 10^{-20} \text{ N s}$ $E = (1467.25 - 1176.36) \times 10^{-12} = 290.86 \times 10^{-12} \text{ J}$	A1 A1 A1 A1																																			
	(e)(ii)2	$m = \sqrt{\frac{E^2 - (p_x^2 + p_y^2 + p_z^2)c^2}{c^4}}$ $= \sqrt{\frac{(290.87 \times 10^{-12})^2 - ((79.46 \times 10^{-20})^2 + (5.77 \times 10^{-20})^2 + (4.83 \times 10^{-20})^2)c^2}{c^4}}$ $= 1.835 \times 10^{-27} \text{ kg}$	C1 A1																																			